You can define alarms on many different metrics. You'll find an overview of all namespaces, dimensions, and metrics that AWS offers at http://mng.bz/8E0X. For example, you could scale based on the load balancer's metric counting the number of requests per target, or the networking throughput of your EC2 instances. You can also publish custom metrics—for example, metrics directly from your application like thread pool usage, processing times, or user sessions.

Scaling based on CPU load with VMs that offer burstable performance

Some virtual machines, such as instance families `t2` and `t3`, offer burstable performance. These virtual machines offer a baseline CPU performance and can burst performance for a short time based on credits. If all credits are spent, the instance operates at the baseline. For a `t2.micro` instance, baseline performance is 10% of the performance of the underlying physical CPU.

Using virtual machines with burstable performance can help you react to load spikes. You save credits in times of low load and spend credits to burst performance in times of high load. But scaling the number of virtual machines with burstable performance based on CPU load is tricky because your scaling strategy must take into account whether your instances have enough credits to burst performance. Consider searching for another metric to scale (such as number of sessions) or using an instance type without burstable performance.

You've now learned how to use autoscaling to adapt the number of virtual machines to the workload. It's time to bring this into action.

17.3 Decoupling your dynamic EC2 instance pool

If you need to scale the number of virtual machines running your blogging platform based on demand, Auto Scaling groups can help you provide the right number of uniform virtual machines, and scaling schedules or CloudWatch alarms can increase or decrease the desired number of EC2 instances automatically. But how can users reach the EC2 instances in the pool to browse the articles you're hosting? Where should the HTTP request be routed?

Chapter 14 introduced the concept of decoupling: synchronous decoupling with ELB and asynchronous decoupling with SQS. If you want to use autoscaling to grow and